

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

PRACTICAL V

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Class: TYIT	Batch: 1
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A:- Using a small dataset (e.g., weather or spam dataset), implement a Naive Bayes classifier using scikit-learn.

Code:-

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import GaussianNB

data = {
    'Outlook': ['Sunny', 'Sunny', 'Overcast', 'Rainy', 'Rainy', 'Rainy', 'Overcast'],
    'Temperature': ['Hot', 'Hot', 'Hot', 'Mild', 'Cool', 'Cool', 'Hot'],
    'Humidity': ['High', 'High', 'High', 'High', 'Normal', 'Normal', 'High'],
    'Windy': ['False', 'True', 'False', 'False', 'False', 'True', 'True'],
    'Play': ['No', 'No', 'Yes', 'Yes', 'Yes', 'No', 'Yes']
}

df = pd.DataFrame(data)

le = LabelEncoder()

for col in df.columns:
    df[col] = le.fit_transform(df[col])

X = df[['Outlook', 'Temperature', 'Humidity', 'Windy']]
y = df['Play']

model = GaussianNB()
model.fit(X, y)

test = pd.DataFrame(
    [[2, 0, 0, 1]],
    columns=['Outlook', 'Temperature', 'Humidity', 'Windy']
)

prediction = model.predict(test)
```

```
if prediction[0] == 1:  
    print("Prediction: Yes, we can Play")  
else:  
    print("Prediction: No, we cannot Play")
```

Output:-

```
Prediction: No, we cannot Play
```